

Kamran Heydarov

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SUMMARY

Detail-oriented and driven Chemical Engineering student with a strong foundation in analytical chemistry and data science, currently completing a B.Sc. at Azerbaijan State Oil and Industry University (expected June 2025). Experienced in chemical analysis, chromatographic and spectroscopic techniques, and thermophysical research. Proficient in Python, KNIME, Aspen HYSYS, and statistical data analysis. Successfully led and contributed to several academic and open-source projects in chemical process simulation, thermodynamic modeling, and educational platform development. Passionate about integrating data science with chemical engineering to improve process efficiency, enhance material characterization, and advance sustainable practices.

EDUCATION

Baku, Azerbaijan Graduation Date: Jun 2027	French-Azerbaijani University <i>Master of Science (M.Sc.) in Physical Chemistry/ Chemical Engineering</i>
Baku, Azerbaijan Graduation Date: Jun 2025	Azerbaijan State Oil and Industry University <i>Bachelor of Science (B.Sc.) in Chemical Engineering (Chemical Technologies Faculty)</i>
Baku, Azerbaijan Graduation Date: Jun 2020	Chemistry-Biology Oriented Republican High School <i>Secondary School</i>

WORK EXPERIENCE

Baku, Azerbaijan Jan 2025 - June 2025	SOCAR Carbamide Plant <i>Chemical engineer intern</i> - Learn the synthesis process of ammonia and urea. - Work with PFD and P&ID diagrams. - Participate in the analysis of chemical synthesis products using gas chromatography. - Examine various physical properties of urea. - Complete projects with work on ASPEN HYSYS and ASPEN Heat Exchanger
Baku, Azerbaijan Sep 2023 - Feb 2025	Azerbaijan State University of Oil and Industry <i>Lab Assistant Supervised by Dr. Leyla Vazirova (email)</i> - Assisted professors in undergraduate labs at UFAZ and ASOIU. - Strengthened love for teaching and research. - Guided 20+ students per lab session, improving their understanding of experimental concepts and helped boost lab grades. - Taught students statistical methods for chemical analysis, enhancing their data interpretation skills and analytical reasoning.
Baku, Azerbaijan Sep 2024 - Oct 2024	Institute of Microbiology <i>Intern</i> - Conducted 50+ microbiological tests and experiments, mastering key techniques such as microbial isolation and culture preparation. - Performed 10+ microbial culture experiments, accurately identifying strains with 95% success using advanced instrumentation. - Utilized advanced instrumentation for analysis. - Prepared and maintained culture media, performed assays, and analyzed experimental data.

Baku, Azerbaijan

Aug 2023 - May 2024

SPE Energy School*Renewable Energy Student*

- Engaged in a 9-month educational initiative, contributing to 5 case studies and 2 group projects on renewable energy solutions.
- Learned theoretical knowledge and honed practical skills through case studies, debates, workshops, projects, and internships.

Baku, Azerbaijan

Aug 2023 - Jan 2024

Institute of Petrochemical Processes*Intern on Petroleum Education*

- Earned experience in oil research through hands-on work in specialized labs.
- Focused on catalysis in oil exploration and refining.
- Received mentorship that fueled passion for a career in petrochemicals.

PROJECT EXPERIENCE

Baku, Azerbaijan

March 2025 - Present

pyionics python library ([link](#))*owner & developer*

- Developed a Python-based tool for thermodynamic property estimation of ionic liquids and mixtures.
- Integrated models for phase equilibrium and heat capacity prediction to support research and education.
- Optimized the codebase for usability in academic and industrial thermodynamic analysis.

Baku, Azerbaijan

March 2025 - Present

Heat Exchanger Design Software*owner & developer*

- Designed and simulated heat exchangers using tools like ChemGraph to analyze thermal performance.
- Applied heat transfer principles to develop efficient systems for academic and research use.
- Helped enhance ChemGraph's features to support heat exchanger modeling.

Baku, Azerbaijan

Feb 2025 - Present

ChemGraph - Project Simulation Software*Executive Director*

- Lead the development of ChemGraph, an open-source PFD design simulator supported by Azerbaijan State Oil and Industry University.
- Oversee its application in education and research, aiding students and researchers in process simulation.
- Enhance the platform's functionality to improve accessibility and usability for academic and industrial applications.

Baku, Azerbaijan

Sep 2024 - Present

ILThermoGUI*Data Scientist (Group of Shamkhal Baybakev)* ([email](#))

- Developed a Python library for installing and accessing NIST ILThermo data.
- Cleaned, processed, and structured datasets for improved usability.
- Designed visualizations to analyze and interpret thermophysical data.
- Conducted statistical analysis to extract meaningful insights.
- Researched ionic liquids, focusing on their thermophysical properties and applications.

Baku, Azerbaijan

Jan 2024 - Present

MentorCorp*Co-Founder & CTO*

- Developed a user-friendly interface on the mentoring platform. Streamlined resource access.
- Achieved an average onboarding time reduction of 30% for new users seeking guidance in professional growth.

Baku, Azerbaijan

Jan 2024 - Sep 2024

YOUNGH*Founder & CEO*

- Built a Q&A platform with 200+ active users, enabling students to earn up to \$500 collectively in its first 6 months.
- Enabled event tracking and provided opportunities through the website.
- Offers a comprehensive platform for academic and professional growth.
- Sold the platform to a company in our competition, expanding its reach and impact.

Baku, Azerbaijan

Jan 2022 - Jan 2023

EvoEco*Co-Founder & CEO*

- Worked on addressing energy challenges through various approaches such as synthesizing energy via algae photosynthesis.
- Developed energy sources for various forms of use.

CERTIFICATIONS

Data Manipulation with pandas | ELA982 SChE® Managing Lab Process Safety 1 | ELA986 SChE® Managing Lab Process Safety 2 | ELA954 SChE® Certificate Program - Lab Safety | MongoDB Python Developer Path | Excel Fundamentals | GitHub Foundations | KNIME Fundamentals | JavaScript (intermediate) Certificate | intermediate Python

LEADERSHIP EXPERIENCE

Baku, Azerbaijan

Feb 2024 - Present

Student Sustainability Commitment at ASOIU (link)*Co-chairman*

- Involved in ASOIU's Sustainability Plan based on a broad block of topics.
- Helped start and advance sustainability initiatives at ASOIU.

Baku, Azerbaijan

Aug 2023 - Present

Science Club (link)*Founder and President*

- Created an alternative to volunteer organizations of foreign companies at the university.
- Organized seminars, competitions, and training in chemistry, physics, mathematics, IT, and more.

INCUBATION PROGRAMS

Baku, Azerbaijan

Jan 2024 - May 2024

SABAH.HUB Incubation*Project Owner - Youngh*

- Built a Q&A platform with 200+ active users, enabling students to earn up to \$500 collectively in its first 6 months.
- Enabled event tracking and provided opportunities through the website.

Baku, Azerbaijan

Oct 2023 - July 2024

UFAZ - IGNITE Incubation Program*Project Owner - Youngh*

- Offers a comprehensive platform for academic and professional growth.
- Sold the platform to a company in our competition, expanding its reach and impact

SKILLS & INTERESTS

Advanced Knowledge: Python (matplotlib, pandas, NumPy, tkinter, rdkit), LATEX, Avogadro, Aspen HYSYS, Aspen Heat Exchanger, KNIME, Excel, MongoDB, Density Function Theory for calculation molecular quantum dynamics

Basic Knowledge: Matlab, CAD (SolidWorks, AUTOCAD)

Languages: Azerbaijani: Native, English: Fluent, Turkish: Native, Russian: B1